

controller in conjunction with many other modules.

Power flow and energy management. The system has three sources of power. First and foremost are solar panels. Sizing estimation is beyond the scope of this article, but it is typically a few kilowatts at the minimum.

A panel typically produces at rated irradiance about 150W/sqm. Solar panels feed MPPT module. This is a DC-DC converter with a maximum power point tracking algorithm running inside it. These are usually very high efficiency units, running at more than 98 per cent electrical efficiency. These are typically multi-phase interleaved buck or buck-boost converters, whose operating levels are at a few hundred volts at both input and output sides.

Isolation may or may not be a requirement, but most implementations are galvanically-isolated for regulatory and safety reasons. Output feeds a common DC bus, from which downstream energy may be provided to the load. Implementation may be analogue, fully digital, or a mix of analogue and digital control.

The second source is the grid. This may be optional, as the intent is to maximise usage of solar. However, in areas where intermittent grid is available, or where solar insolation is not entirely sufficient for operation year long, or during certain seasons, the grid helps in fulfilling the demand. Since the system is essentially a solar energy storage setup, it is also possible to use this station to supplement the grid during peak hours or as a solar farm, using bi-directional grid-tied inverters. With

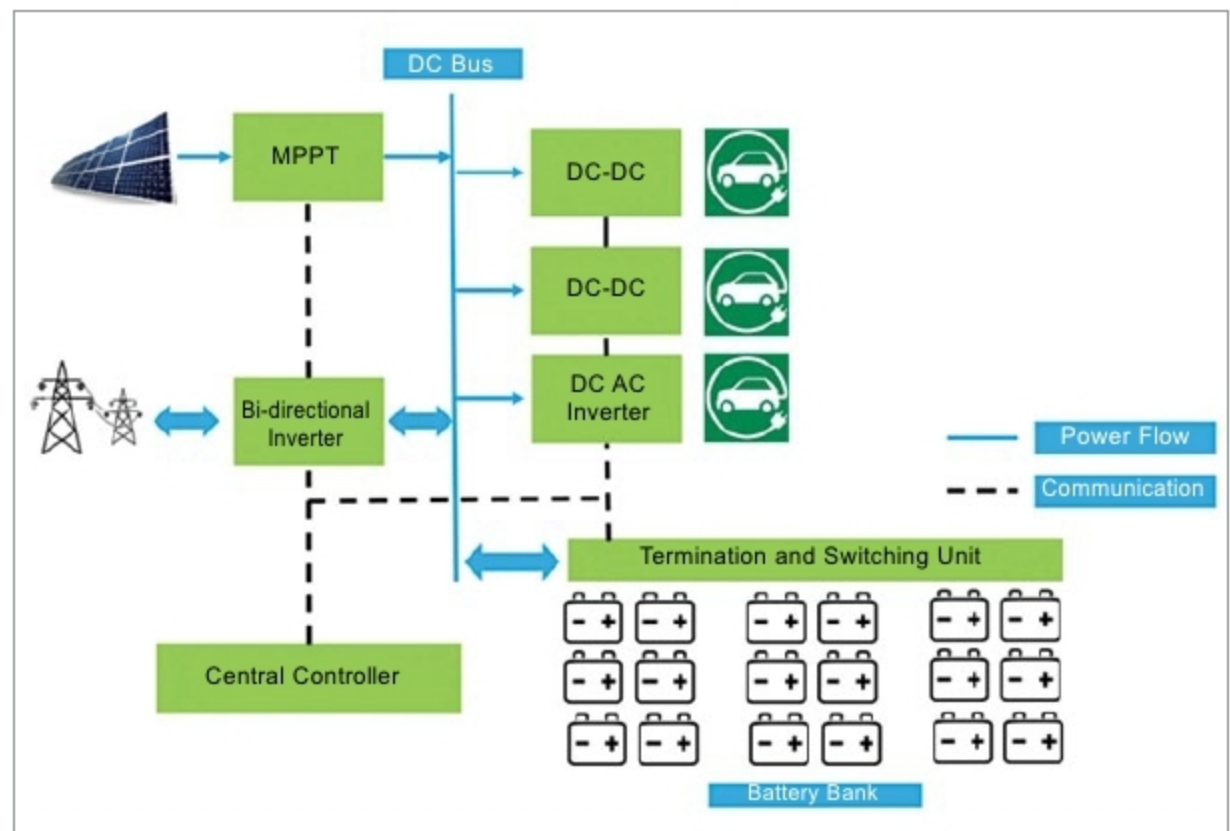


Fig. 2: Power system architecture at the backend

proper policies in place for exporting to the grid from solar farms or captive plants with net-metering, this serves a dual purpose, too.

The third source or sink/storage is the battery. The trend these days is to use lithium-ion batteries that have a very high cycle life, lend themselves well to quick charging, have very high depth of discharge and very high volumetric efficiency. It is possible to house these batteries underground to save real estate.

Lithium-ion battery packs are arranged in a suitable series parallel combination, and in several strings. The batteries terminate themselves into a junction box and termination unit, which also functions as a supervisor.

Each battery has a data port, typically CAN or RS485, and these are daisy chained and fed to this termina-

tion unit, which then has a top-level view of the health and status of every individual battery, string or the entire battery bank. This is essentially a data concentrator and a switching unit, putting battery packs in or out of circuit. In addition, this communicates with the central controller to decide charge and discharge of the batteries.

Fig. 2 makes the power system architecture quite clear. This is a modular system to allow for suitable expansion, and the 3kW to 5kW modules are typically expandable with a communication bus, typically CAN or MODBUS/RS485. The central controller is able to configure the modules as per functional requirement at any point of time—be it charge management, load management or diagnostic checks.

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